Vibe Research: Framework, Trust, and Collaboration

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Vibe Coding Is Transforming Software Development

- Developer productivity has surged
- People with no coding experience can build apps with AI
- Validation is simple and immediate: if it runs, it works

This trend is spreading into scientific research —

Vibe Research.

But Science Is Different

Vibe Coding

- Primarily an **individual** activity
- Write a program; if it runs, you're done
- Clear validation criteria

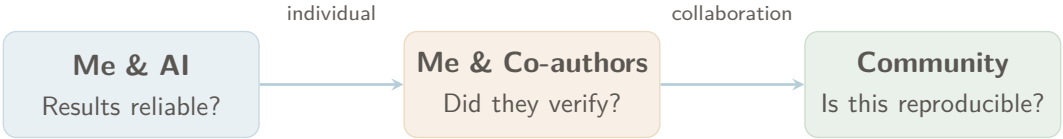
Scientific Research

- Inherently **collaborative**
- Results must be **trustworthy**
- Methods must be **rigorous**
- Conclusions must withstand **peer replication**

A working app and a reliable scientific conclusion are **entirely different things**.

Vibe Research methods are still immature; tools are designed for **individuals**. This creates a fundamental tension with the social structure of science.

The Core Challenge: Trust



These three layers of trust differ in nature and require different strategies.

This talk unfolds along these three dimensions.

Paper as Promise: A Trust Framework

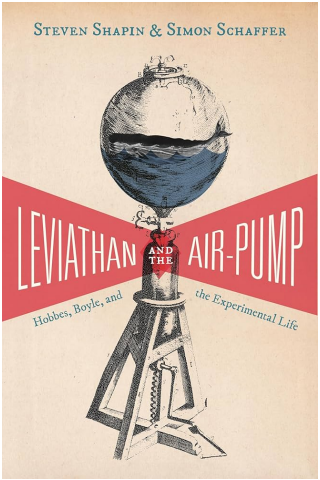
Historical Origins: Virtual Witnessing

What are scientists actually doing? What *is* a paper?

17th Century: Boyle's Air-Pump Experiments

- Most people could not witness experiments in person
- Boyle's solution: a "literary technology"
- Extraordinarily detailed experimental reports
- Creating **virtual witnessing** for the reader
- As if they had seen it with their own eyes

— Shapin & Schaffer, *Leviathan and the Air-Pump*



Shapin & Schaffer

Leviathan and the Air-Pump

From Witnessing to Promise

Early science: "virtual witnessing" = making readers believe "I saw this"

Modern papers go further: a paper is a **promise**.

It promises:

> *If you follow my method,*
> *you will see my reported results.*

- Early science: "witnessing" → trust
- Modern science: "**reproduction**" → trust

Popper's falsifiability = realized through the paper as **social contract**

Karl Popper

(1902–1994)

The “Promise” Structure of a Paper

Structure of a Scientific Paper

Background What / Why?

Method What did we do?

Math principles & algorithms

Results What did we observe?

Reproducible experimental data

Promise

“Follow this method; you will see these results”

From Human Promise to Human + AI Promise

Traditionally, a paper's promise is backed by the **researcher's competence and integrity**.

In Vibe Research, much of the work is done by AI:

- The **guarantee structure** of the promise has fundamentally changed
- Are AI-generated results reliable?
- The key question is not "yes or no" but: **how do you verify?**

This leads to the workflow I've developed through practice.

Individual Level:
Trust Between Me and AI

How I View AI

Core Principle

AI = **"A talented but untrustworthy remote research team"**

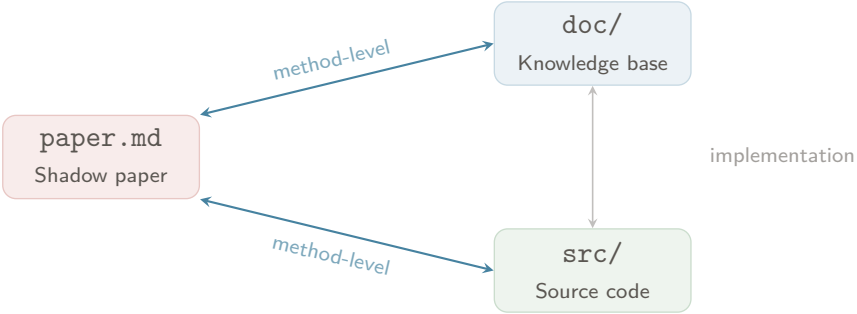
- Highly capable, massive output
- But you **cannot blindly trust** anything it produces
- Human's role: **set direction**, **review results**
- Interact through a **compressed document interface**

Two Agent Skills* built for this:

- **Vibe Research Skill**
  Research workflow
  `github.com/Zhangyanbo/vibe-research-skill`
- **Vibe Paper Writing Skill**
  Paper writing
  `github.com/Zhangyanbo/vibe-paper-writing`

*Agent Skill: a skill module loaded into AI coding assistants that constrains AI behavior via predefined rules.

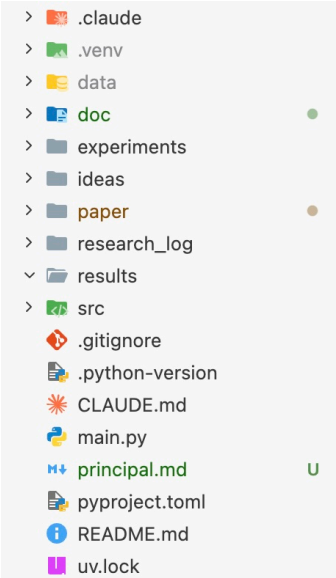
Bi-directional Sync: The Soul of the System



Highest-Priority Rule — Never Skip

Code changed (algorithm, loss function, architecture...) → `paper.md` must sync
`paper.md` method section edited → code must adapt accordingly

In Practice: A Real Project



Key folders in a Vibe Research project:

- `paper/` — the synchronized shadow paper
- `doc/` — experiment workflow & design decisions
  So a new Claude Code session doesn't need to re-read all code
- `research_log/` — append-only session reports
- `src/` — source code
- `results/` — experiment outputs

These three surfaces (`paper/`, `doc/`, `src/`) stay in sync at all times.

`paper.md`: A Compressed, Falsifiable Commitment

`paper.md` is not passive documentation —

- It doesn't just describe what code **does**; it declares what code **should produce**
- The "paper as promise" framework **implemented in engineering**

Strict constraints:

- Only four sections: Abstract, Introduction, Method, Results
- Method: **only** mathematical principles and algorithmic frameworks
- No class names, function calls, or file paths allowed

→ Forces documentation to stay at the **algorithmic abstraction layer**

Why "Paper" Instead of "Code"?

Core idea: **Don't read the code — focus on the math and algorithms in the paper**

Avoiding the Bottleneck

AI output is massive
Line-by-line review → human bottleneck

Paper = higher abstraction layer
Verify at the methodology level without drowning in implementation

Compression = Foundation of Collaboration

A good `paper.md`
= **compressed representation** of the project

Anyone can quickly understand: what, how, and what results

No need to read thousands of lines

Reproducibility: Verifying the Promise

Verification workflow:

1. Extract `paper.md` in isolation
2. Hand it to a **fresh AI**
3. No access to original code, data, or docs
4. Re-implement from scratch
5. Results match $\rightarrow$ promise verified

= Simulating **peer replication** in your own lab

Theory: independent replication lets "hallucinations" cancel out statistically

Independence: The Mathematical Limit of Trust

This approach has a fundamental limitation: the **independence problem**.

Traditional replication:

- Lab **independence** = source of trust

Between two AIs:

- May share the same biases
- "Hallucination cancellation" assumes **statistical independence**
- This assumption **often doesn't hold**

Mathematically:

- High model correlation → **diminishing returns**
- Accuracy has an **upper bound**

Practical advice:

- Use **different** AI models
- Human judgment remains **irreplaceable**
- Read `paper.md` yourself

Vibe Paper Writing Skill

Handles the other direction: converting **informal input** into **rigorous academic text**

Input: chat logs, emails, notes → Output: LaTeX paper

github.com/Zhangyanbo/vibe-paper-writing

Three principles:

1. **Endorsement Awareness** — only incorporate explicitly approved content
2. **Voice Preservation** — maintain the paper's established register and style
3. **Compilation-Gated** — no edit is complete until LaTeX compiles cleanly

The Expanding Boundaries of Vibe Research

- **Paper reading**: arXiv papers have LaTeX source → AI translates instantly
- **Hypothesis generation**: human-AI dialogue; hard to fully automate

A Long-Term View

Specific technologies don't matter — AI will replace all technical tools.

The real question: **how will AI change the scientific community itself?**

Collaboration Level: When Production Outpaces Trust

Baumol's Cost Disease

Economist William Baumol, 1960s:

- Some sectors see **massive productivity gains** (manufacturing)
- Others are **inherently resistant** to productivity growth

Classic example: **the string quartet**

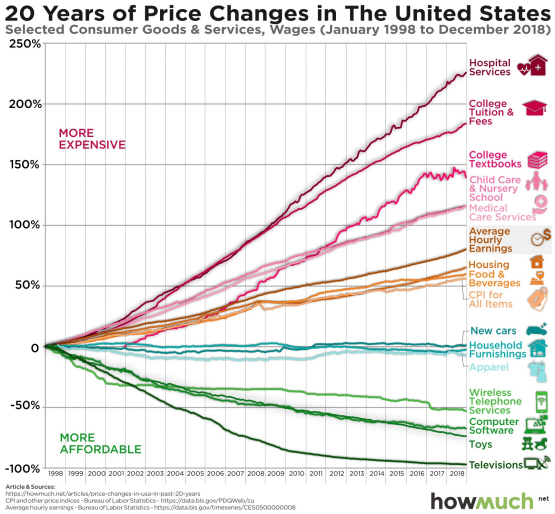
A Mozart piece required 4 musicians for 40 minutes — 200 years ago, and still does today.

Wages must rise with the economy
→ "stagnant" sectors become relatively more expensive
→ Education, healthcare, live performance: prices **always outpace inflation**

William Baumol

(1922–2017)

Baumol's Cost Disease: The Data



US price changes 1998–2018. Source: Wikipedia

Labor-intensive services

Healthcare, education ↑ 150–250%

Technology-driven manufacturing

TVs, software, toys ↓ 50–100%

Economics offers **no easy solutions**:

1. Accept it
2. Boost productivity in stagnant sectors
3. Redesign processes

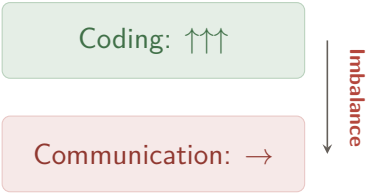
Baumol's Cost Disease in Vibe Research

The same dynamic is unfolding:

Coding and experimentation
→ productivity soaring

Communication, understanding, trust-building
→ productivity nearly unchanged

Communication and trust may be like the string quartet —
inherently resistant to automation
— because trust requires **human investment and time**



AI accelerates divergence,
but not consensus.

The Communication Ratio Imbalance

≈ balanced

Traditional | Coding | Communication & trust

Imbalance

Vibe Coding | Coding (AI ×10) | Communication & trust

Production cost → **near zero**

Communication cost → **unchanged**

Conflicts Are Not a Tooling Problem

Git example:

- Two people push fast on separate branches
- Divergence grows far faster than before
- Merge conflicts become extremely hard to resolve

The **root cause** is not inadequate tools:

Key Insight

You often don't know what you want
until you see what someone else built.

→ We need collaboration that **eliminates conflicts by design**.

The Shift in Tacit Knowledge

Michael Polanyi: "We know more than we can tell"

Traditional: reading code together, line-by-line discussion, whiteboard debates
→ accumulating **shared tacit knowledge**

After Vibe Research — an unsettling situation:

- You don't fully know **your own** code (AI wrote it)
- Your collaborator doesn't fully know **their** code either
- You know **their** code even less

`paper.md` with bi-directional sync solves the **individual** problem. But in **collaboration**: how do two independent `paper.md` files merge? — *no good solution exists yet*.

The Trust Crisis in Collaboration

Before: collaborator produces slowly → more time to think → lower error rate; you can verify

Now: AI lets the same person produce massive output quickly; unreliable content likely mixed in

Core issue: **procedural opacity**

- Did they use a good enough model? (free tier vs. best paid — enormous gap)
- Did they verify? Was their prompting strategy sound?
- **You cannot know how they used AI**

Skepticism about the technology ⟶ amplifies distrust of the person

Community Level: Challenges for the Scientific Community

The Tension Between Senior and Junior Researchers

AI appears to "democratize" research ability — but the data tells a more complex story.

Toner-Rodgers (2024): randomized experiment with 1,018 scientists

- AI-assisted → 44% more new materials discovered
- But effects **vary dramatically** by ability:
  - Bottom 1/3: almost no benefit
  - Top 10%: output up 81%
- AI automated "idea generation" tasks
- Scarcest skill shifted from "creativity" to **"judgment"**

Core Conclusion

The market doesn't need more "people who can produce with AI"

It needs more "**people who can judge whether AI output is reliable**"

AI didn't democratize research — **it widened the judgment gap**

The PhD Student Growth Paradox

Becoming a senior researcher requires:

- Independent judgment, methodological intuition, error sensitivity
- Built through "**working slowly, failing, gradually understanding**"

AI skips this process.

Exoskeleton athlete — great times, but muscles never developed.

Time-delay effect:

- Today: more papers, faster graduation
- 5 years later: can they judge reliability independently?
- Risk: "**high-output, low-judgment**" generation

Peer Review and Credit Assignment

As AI involvement in papers grows:

Peer review:

- You see a paper; you know AI was likely involved
- But you don't know how much the author verified
- You don't know if they used AI responsibly
- Under this **information asymmetry** — do you trust it or not?

Credit assignment:

- Who is first author? Who contributed the core idea?
- When AI wrote much of the code and even parts of the analysis
- Do traditional contribution metrics still apply?

The Road Ahead

“One-Person Science” and New Forms of Collaboration

If trust problems remain unsolved

- AI gives individuals immense capability
- Solo research **increasingly feasible**

However —

- Science moved toward large-scale collaboration for good reasons
- Problem **complexity keeps growing**

If collaboration changes form

- No more “shared codebase, co-editing one paper”
- More like **microservices**:
  - Each person’s “promise” module
  - Clear interface contracts

However —

- Innovation emerges at **blurry, cross-disciplinary boundaries**
- Hardest place to define interfaces

Three Paths Forward (Baumol's Lens)

1. **Accept rising communication costs**
   Communication and verification take ever more time — the new normal
2. **Improve communication & trust efficiency** $\leftarrow$
   "Paper as promise" + bi-directional sync = compress communication costs
3. **Redesign collaboration architecture** $\leftarrow$
   From labs to communities — adapt to AI-era production rhythms

Passively accepting rising costs $\rightarrow$ **humans become the bottleneck**
— exactly the problem Vibe Research aims to solve.

Scientific Taste and Case Law: An Analogy

How can AI develop a scientist's "taste" — judging which directions are worth pursuing?

Civil law: encode all rules in statutes

- Like writing all requirements into a prompt
- But novel situations always arise

Common law: rely on precedent

- Facing a new case, search past precedents
- Rule based on the closest match

Scientific "taste" $\approx$ **precedent** — the micro-decisions a senior researcher makes (which direction to pursue, which to abandon) are often tacit, and scientists habitually **delete the exploration process** from papers.

E.g., Feynman hid his reasoning paths — appearing brilliant, but preventing others from seeing his taste.

(Wolfram, *Idea Makers*)

Proposal: A Scientific Precedent Database

Idea: record micro-decisions → **precedent database** for AI

Like legal "distinguishing precedents":

- Not just finding similar cases
- Judging which dimensions are **similar** vs **different**

Obstacle: manual recording = cognitive burden

Solution: all research within the AI framework

- Every interaction naturally recorded
- Precedents = **byproduct**, not extra work

`research_log/` is a first step, but granularity is still insufficient.

Open Questions

- What kind of **new contracts** do collaborators need to rebuild trust in the Vibe Research era?
- When you see a paper likely involving AI, how do you judge whether the author used AI **responsibly**?
- How do we cultivate the **judgment** the next generation of scientists needs, without giving up the efficiency AI provides?
- How should **peer review** and **credit assignment** evolve for this new paradigm?

These questions have no ready answers.
But only by confronting them can Vibe Research become a force for scientific progress.

Thank You!

Questions & Discussion Welcome